phsdymp2b

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[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Genres I have chosen:

1. **Action**
2. **Comedy**
3. **Animation**
4. **Biography**
5. **Drama**

Reading all three Datasets

[4]:
```
DataA = pd.read_csv('IMDb_movies/IMDb_Dataset_1.csv').drop_duplicates()
Data1 = DataA[(DataA["Genre"] == "Action") |
         (DataA["Genre"] == "Comedy") |
         (DataA["Genre"] == "Animation") |
         (DataA["Genre"] == "Biography") |
         (DataA["Genre"] == "Drama")]
Data1
```

[4]:
```
                                          Title  IMDb Rating  Year  \
0                                     Gladiator          8.5  2000
1    Mission: Impossible - Dead Reckoning Part One          7.7  2023
2             Rebel Moon - Part Two: The Scargiver          5.2  2024
4                                   Borderlands          7.1  2024
5                                 Jurassic Park          8.2  1993
..                                            …          …   …
391                               Anyone But You          6.1  2023
```

```
394                              Kinds of Kindness          7.0  2024
395                            Mother of the Bride          4.8  2024
397                                The Idea of You          6.4  2024
399                                   The Fall Guy          7.1  2024

    Certificates   Genre               Director  \
0              R  Action           Ridley Scott
1          PG-13  Action  Christopher McQuarrie
2          PG-13  Action            Zack Snyder
4              R  Action               Eli Roth
5          PG-13  Action       Steven Spielberg
..           …     …                    …
391            R  Comedy             Will Gluck
394            R  Comedy       Yorgos Lanthimos
395        TV-PG  Comedy            Mark Waters
397            R  Comedy      Michael Showalter
399        PG-13  Action           David Leitch

                                       Star Cast  MetaScore  \
0       David FranzoniJohn LoganWilliam Nicholson       67.0
1              Tom CruiseHayley AtwellVing Rhames       81.0
2            Zack SnyderKurt JohnstadShay Hatten       35.0
4                            Eli RothJoe Crombie       66.9
5                    Michael CrichtonDavid Koepp       68.0
..                                             …        …
391                       Ilana WolpertWill Gluck       52.0
394             Efthimis FilippouYorgos Lanthimos       72.0
395  Brooke ShieldsMiranda CosgroveBenjamin Bratt       39.0
397     Anne HathawayNicholas GalitzineElla Rubin       67.0
399   Ryan GoslingEmily BluntAaron Taylor-Johnson       73.0

     Duration (minutes)
0                 155.0
1                 163.0
2                 122.0
4                 116.3
5                 127.0
..                  …
391               103.0
394               164.0
395                88.0
397               115.0
399               126.0

[181 rows x 9 columns]
```

DataSet 1 contains 181 rows and 9 columns of Genres Action, comedy , Animation , Biography

and Drama

[6]:
```
DataB = pd.read_csv('IMDb_movies/IMDb_Dataset_2.csv').drop_duplicates()
Data2 = DataB[(DataB["Genre"] == "Action") |
         (DataB["Genre"] == "Comedy") |
         (DataB["Genre"] == "Animation") |
         (DataB["Genre"] == "Biography") |
         (DataB["Genre"] == "Drama")]
Data2
```

[6]:
```
                        Title  IMDb Rating  Year Certificates      Genre  \
1               Elvira Madigan          7.0  1967           PG  Biography
4                       Mahler          7.0  1974           PG  Biography
5               The Dark Horse          7.4  2014            R  Biography
7                   Carrington          6.8  1995            R  Biography
8                       Burden          6.7  2018            R  Biography
…                            …          …    …           …         …
3160                Fight Club          8.8  1999            R      Drama
3161              Forrest Gump          8.8  1994        PG-13      Drama
3163                 12th Fail          8.9  2023            R  Biography
3170           The Dark Knight          9.0  2008        PG-13     Action
3172  The Shawshank Redemption          9.3  1994            R      Drama

                    Director  \
1               Bo Widerberg
4                Ken Russell
5     James Napier Robertson
7        Christopher Hampton
8             Andrew Heckler
…                        …
3160           David Fincher
3161         Robert Zemeckis
3163      Vidhu Vinod Chopra
3170       Christopher Nolan
3172          Frank Darabont

                                          Star Cast  MetaScore  \
1           Pia DegermarkThommy BerggrenLennart Malmer         66
4               Robert PowellGeorgina HaleLee Montague         66
5             Cliff CurtisJames RollestonKirk Torrance         77
7         Emma ThompsonJonathan PryceSteven Waddington         66
8     Garrett HedlundAndrea RiseboroughForest Whitaker         57
…                                                  …         …
3160                   Brad PittEdward NortonMeat Loaf         67
3161                  Tom HanksRobin WrightGary Sinise         82
3163    Vidhu Vinod ChopraJaskunwar KohliAnurag Pathak         66
3170     Jonathan NolanChristopher NolanDavid S. Goyer         84
3172               Tim RobbinsMorgan FreemanBob Gunton         82
```

```
      Duration (minutes)
1                   91.0
4                  115.0
5                  124.0
7                  121.0
8                  117.0
…                  …
3160               139.0
3161               142.0
3163               116.3
3170               152.0
3172               142.0

[1969 rows x 9 columns]
```

DataSet 2 contains 1969 rows and 9 columns of Genres Action, comedy , Animation , Biography and Drama

[8]:
```
DataC = pd.read_csv('IMDb_movies/IMDb_Dataset_3.csv').drop_duplicates()
Data3 = DataC[(DataC["Genre"] == "Action") |
         (DataC["Genre"] == "Comedy") |
         (DataC["Genre"] == "Animation") |
         (DataC["Genre"] == "Biography") |
         (DataC["Genre"] == "Drama")]
Data3
```

[8]:
```
                            Title         Director         Star Cast  Year  \
1           Pokémon the Movie 2000  Kunihiko Yuyama   Veronica Taylor  1999
2    When Hitler Stole Pink Rabbit    Caroline Link  Riva Krymalowski  2019
4                     Summer Magic    James Neilson      Hayley Mills  1963
6        The Powerpuff Girls Movie  Craig McCracken    Cathy Cavadini  2002
7                     Camp Hideout       Sean Olson       Corbin Bleu  2023
..                             …              …               …    …
995          The Boy and the Heron   Hayao Miyazaki      Soma Santoki  2023
996                Despicable Me 4     Chris Renaud      Steve Carell  2024
997                     Inside Out      Pete Docter       Amy Poehler  2015
998                             IF   John Krasinski    Cailey Fleming  2024
999                   Inside Out 2      Kelsey Mann       Amy Poehler  2024

     IMDb Rating  MetaScore Certificates      Genre  Duration (minutes)
1            6.1       28.0            G  Animation                  99
2            7.0       54.0           PG  Biography                 100
4            6.9       57.1            G     Comedy                 110
6            6.5       65.0           PG  Animation                  73
7            4.4       57.1           PG     Comedy                 100
..           …         …           …         …                  …
```

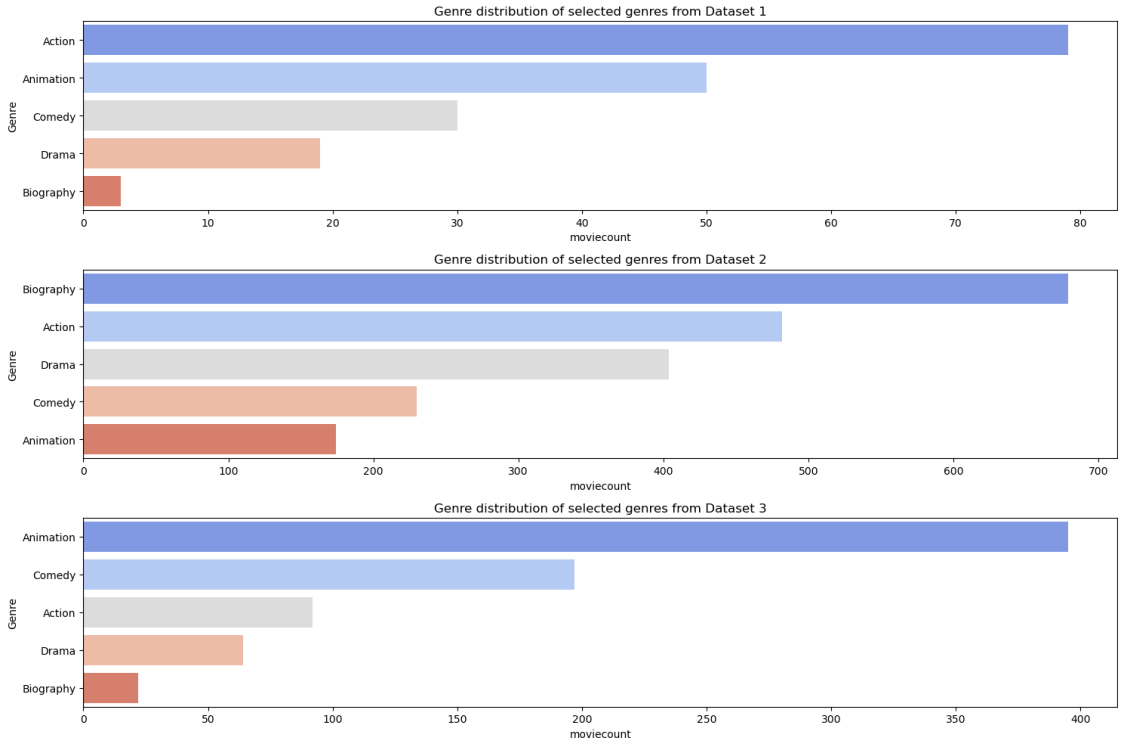
```
995          7.5       91.0        PG-13  Animation                 124
996          6.3       50.0           PG  Animation                  95
997          8.1       94.0           PG  Animation                  95
998          6.5       46.0           PG  Animation                 104
999          7.9       73.0           PG  Animation                  96

[770 rows x 9 columns]
```

DataSet 3 contains 770 rows and 9 columns of Genres Action, comedy , Animation , Biography and Drama

1.1 1. Perform data exploration of movies, genre, and rating and describe the datasets (2 M)

[11]:
```
D1Genre= Data1["Genre"].value_counts().reset_index(name = "moviecount")
D2Genre= Data2["Genre"].value_counts().reset_index(name = "moviecount")
D3Genre= Data3["Genre"].value_counts().reset_index(name = "moviecount")
```

[12]:
```
fig, axes = plt.subplots(3 ,1 , figsize=(15, 10))
sns.barplot(data=D1Genre, y="Genre", x="moviecount",hue="Genre",legend=False,␣
 ↪palette="coolwarm",ax=axes[0])
sns.barplot(data=D2Genre, y="Genre", x="moviecount",hue="Genre",legend=False,␣
 ↪palette="coolwarm",ax=axes[1])
sns.barplot(data=D3Genre, y="Genre", x="moviecount",hue="Genre",legend=False,␣
 ↪palette="coolwarm",ax=axes[2])
axes[0].set_title("Genre distribution of selected genres from Dataset 1")
axes[1].set_title("Genre distribution of selected genres from Dataset 2")
axes[2].set_title("Genre distribution of selected genres from Dataset 3")
plt.tight_layout()
plt.show()
```



From The above plot we can distinguish the datasets as

DataSet1 : As it has large number of action then followed by animation, The Interpretation is DataSet1 contains all kinds of movies which are popular.

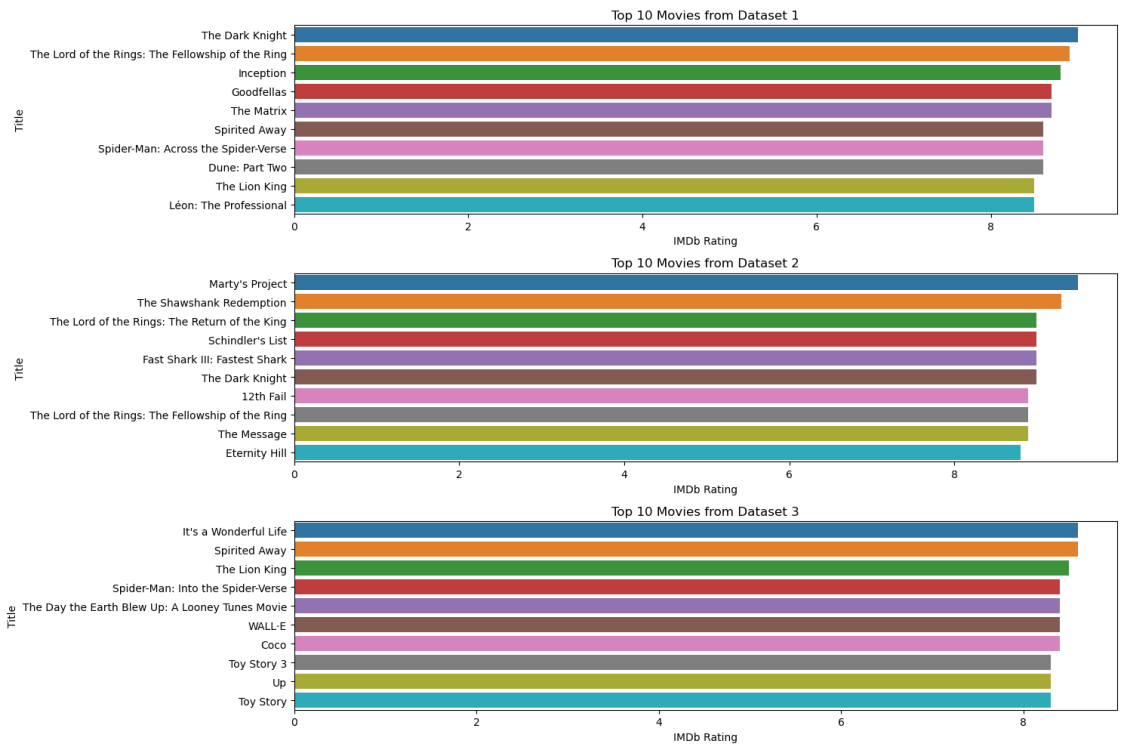
DataSet2 : It contains Biography documentary action adventure crime Horror drama Sci-Fi these kind of movies They mostly have PG, PG-13, or R ratings.(Which says that this genre is mostly for Mature audience who love to watch mostly Biography or Action.)

DataSet3 : While they span different genres (e.g., documentary, animation, biography, adventure, comedy) and target varied age groups, they are all rated G, PG, or PG-13, indicating that they are appropriate for viewing by families, including children

[14]:
```
D1Top10 = Data1.sort_values(by="IMDb Rating",ascending = False).head(10)
D2Top10 = Data2.sort_values(by="IMDb Rating",ascending = False).head(10)
D3Top10 = Data3.sort_values(by="IMDb Rating",ascending = False).head(10)
```

[15]:
```
fig, axes = plt.subplots(3 ,1 , figsize=(15, 10))
sns.barplot(data =D1Top10 , y= "Title",x="IMDb Rating", ax =␣
 ↪axes[0],hue="Title")
sns.barplot(data =D2Top10 , y= "Title",x="IMDb Rating", ax =␣
 ↪axes[1],hue="Title")
```

```
sns.barplot(data =D3Top10 , y= "Title",x="IMDb Rating", ax =␣
↪axes[2],hue="Title")
axes[0].set_title("Top 10 Movies from Dataset 1")
axes[1].set_title("Top 10 Movies from Dataset 2")
axes[2].set_title("Top 10 Movies from Dataset 3")
plt.tight_layout()
plt.show()
```



The Top 10 Movies from DataSet1 are suggested by IMDb if you are a cinema lover

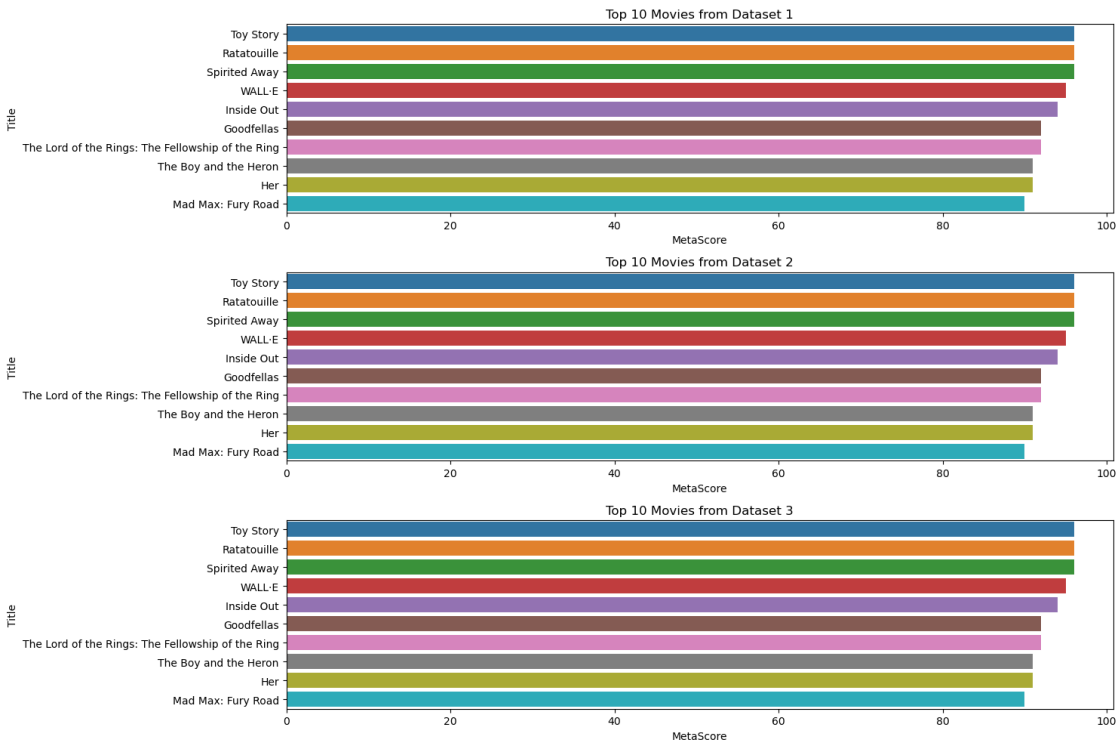
The Top 10 Movies from DataSet2 are suggested by IMDb if you are a person who like biography or action genre

The Top 10 Movies from DataSet1 are suggested by IMDb if you want to watch a Family friendly movie

[17]:
```
Meta1 = Data1.sort_values(by="MetaScore",ascending = False).head(10)
Meta2 = Data2.sort_values(by="MetaScore",ascending = False).head(10)
Meta3 = Data3.sort_values(by="MetaScore",ascending = False).head(10)
```

[18]:
```
fig, axes = plt.subplots(3 ,1 , figsize=(15, 10))
sns.barplot(data =Meta1 , y= "Title",x="MetaScore", ax = axes[0],hue="Title")
sns.barplot(data =Meta1 , y= "Title",x="MetaScore", ax = axes[1],hue="Title")
```

```
sns.barplot(data =Meta1 , y= "Title",x="MetaScore", ax = axes[2],hue="Title")
axes[0].set_title("Top 10 Movies from Dataset 1")
axes[1].set_title("Top 10 Movies from Dataset 2")
axes[2].set_title("Top 10 Movies from Dataset 3")
plt.tight_layout()
plt.show()
```



The Top 10 Movies from DataSet1 are suggested by MetaScore if you are a cinema lover

The Top 10 Movies from DataSet2 are suggested by MetaScore if you are a person who like biography or action genre

The Top 10 Movies from DataSet1 are suggested by MetaScore if you want to watch a Family friendly movie

1.2 2. Perform the required data integration (from different datasets), cleaning for missing or inconsistent data. (2 M)

[21]:
```
df1 = pd.concat(map(pd.read_csv, ['IMDb_movies/IMDb_Dataset_1.csv',
                                  'IMDb_movies/IMDb_Dataset_2.csv',
                                  'IMDb_movies/IMDb_Dataset_3.csv']),␣
 ↪ignore_index=True)
```

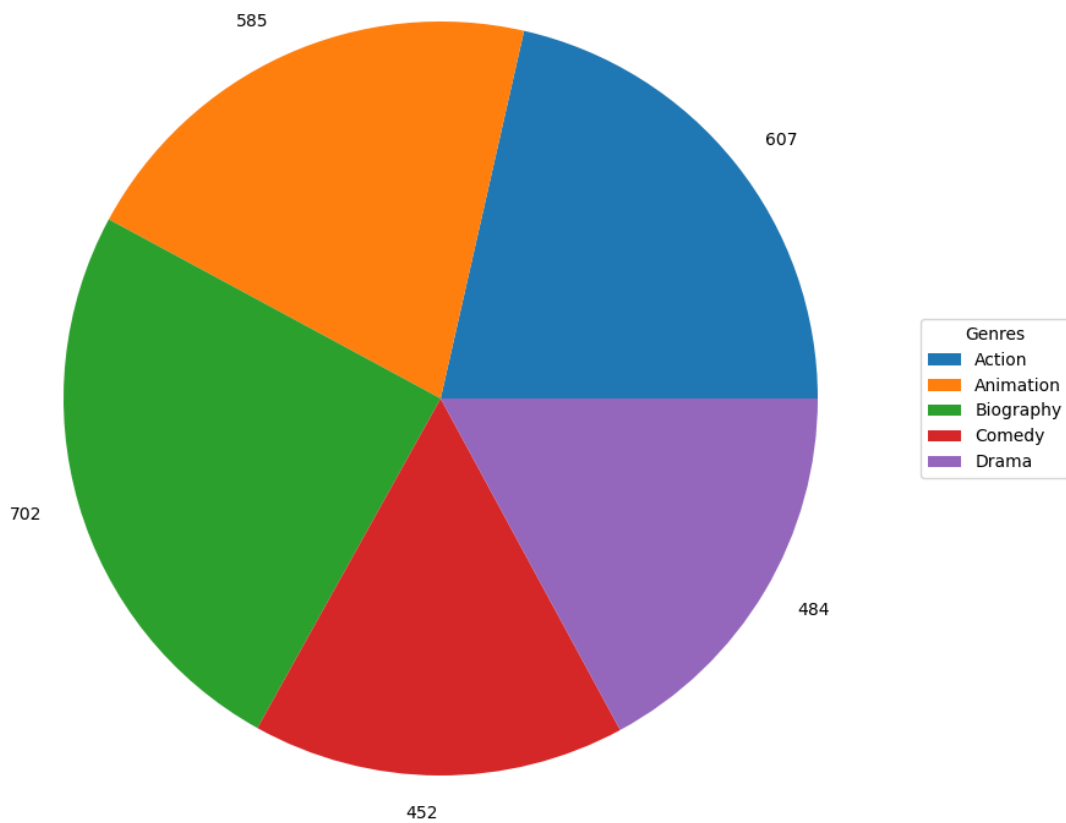
[22]:
```
df1 = df1.drop_duplicates()
df = df1[(df1["Genre"] == "Action") |
         (df1["Genre"] == "Comedy") |
         (df1["Genre"] == "Animation") |
         (df1["Genre"] == "Biography") |
         (df1["Genre"] == "Drama")]
df
```

[22]:
```
                                          Title  IMDb Rating  Year  \
0                                      Gladiator          8.5  2000
1     Mission: Impossible - Dead Reckoning Part One          7.7  2023
2              Rebel Moon - Part Two: The Scargiver          5.2  2024
4                                    Borderlands          7.1  2024
5                                  Jurassic Park          8.2  1993
…                                              …          …    …
4568                        The Boy and the Heron          7.5  2023
4569                              Despicable Me 4          6.3  2024
4570                                   Inside Out          8.1  2015
4571                                           IF          6.5  2024
4572                                  Inside Out 2          7.9  2024

     Certificates      Genre               Director  \
0               R     Action           Ridley Scott
1           PG-13     Action  Christopher McQuarrie
2           PG-13     Action            Zack Snyder
4               R     Action               Eli Roth
5           PG-13     Action       Steven Spielberg
…           …        …                    …
4568        PG-13  Animation         Hayao Miyazaki
4569           PG  Animation           Chris Renaud
4570           PG  Animation            Pete Docter
4571           PG  Animation         John Krasinski
4572           PG  Animation            Kelsey Mann

                                     Star Cast  MetaScore  Duration (minutes)
0     David FranzoniJohn LoganWilliam Nicholson       67.0               155.0
1            Tom CruiseHayley AtwellVing Rhames       81.0               163.0
2           Zack SnyderKurt JohnstadShay Hatten       35.0               122.0
4                           Eli RothJoe Crombie       66.9               116.3
5                   Michael CrichtonDavid Koepp       68.0               127.0
…                                             …        …                  …
4568                              Soma Santoki       91.0               124.0
```

```
4569                          Steve Carell       50.0                95.0
4570                            Amy Poehler      94.0                95.0
4571                         Cailey Fleming      46.0               104.0
4572                            Amy Poehler      73.0                96.0

[2830 rows x 9 columns]
```

1.3 3. Use Python and libraries like Pandas, Matplotlib and Seaborn to create insightful tabulations and visualization for the following: (6M)

[24]:
```
fig, axes = plt.subplots(1 ,1 , figsize=(10, 10))
Genre = df.groupby("Genre")["Genre"].count().reset_index(name="MovieCount")
plt.pie(Genre["MovieCount"],labels=Genre["MovieCount"])
plt.legend(Genre["Genre"], title="Genres", loc="center left",␣
 ↪bbox_to_anchor=(1, 0.5))
plt.show()
```



This is the Distribution of Data W.R.T Genres chosen

Action - 607

Animation - 585

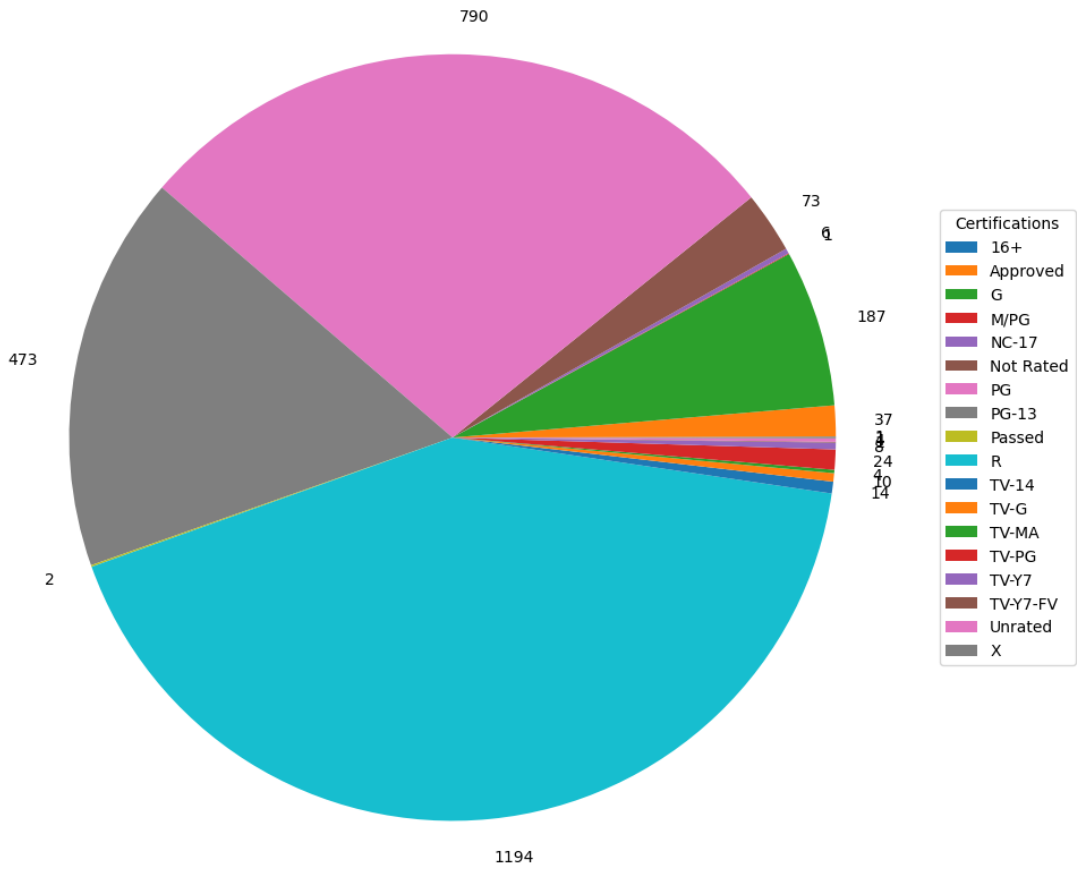
Biography - 702

Comedy - 452

Drama - 484

People are prefering to choose Biography and Action genres from chosen 5 genres which signifies that Makers are more intrested in showing These kind of content to mature audiences

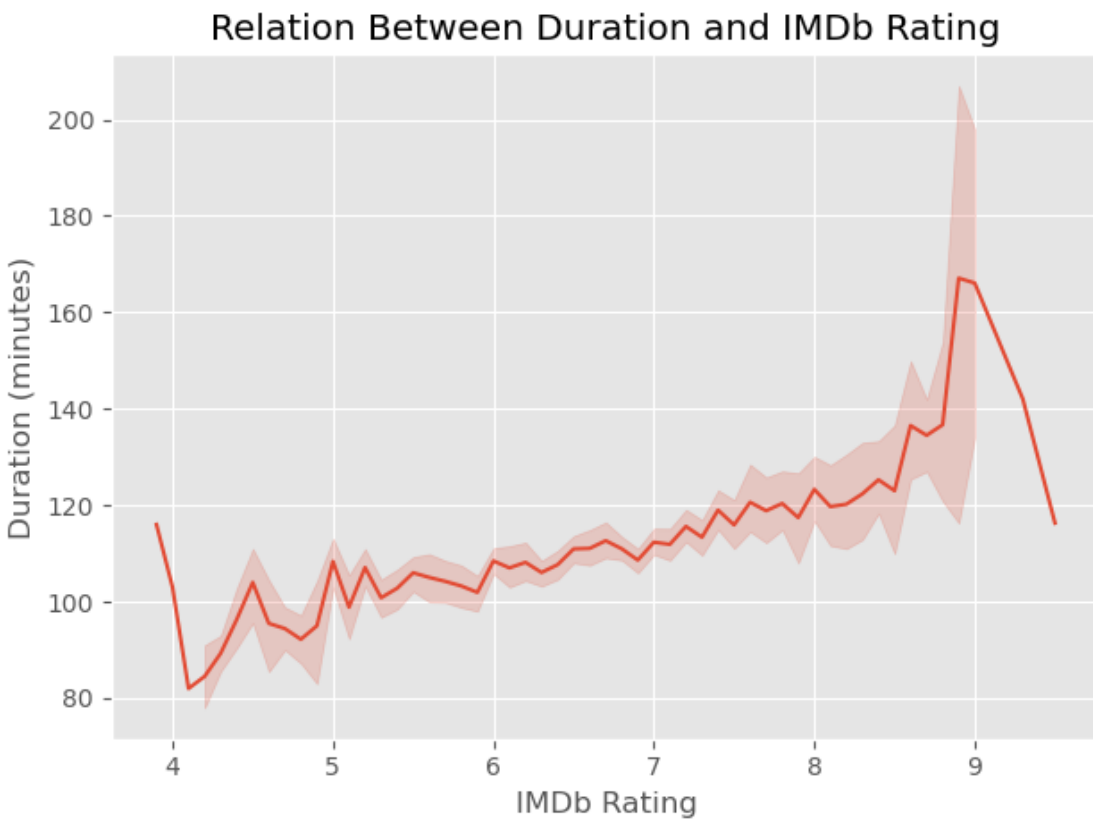
[26]:
```
fig, axes = plt.subplots(1 ,1 , figsize=(10, 10))
Certi = df.groupby("Certificates")["Certificates"].count().
 ↪reset_index(name="MovieCount")
plt.pie(Certi["MovieCount"],labels=Certi["MovieCount"])
plt.legend(Certi["Certificates"], title="Certifications", loc="center left",␣
 ↪bbox_to_anchor=(1, 0.5))
plt.tight_layout()
plt.show()
```



This Pie diagram Shows the Number of R(1194) , PG(790) , PG-13(473) Rated are very high which again proves the fact that movies are more concentrated towards elder audience

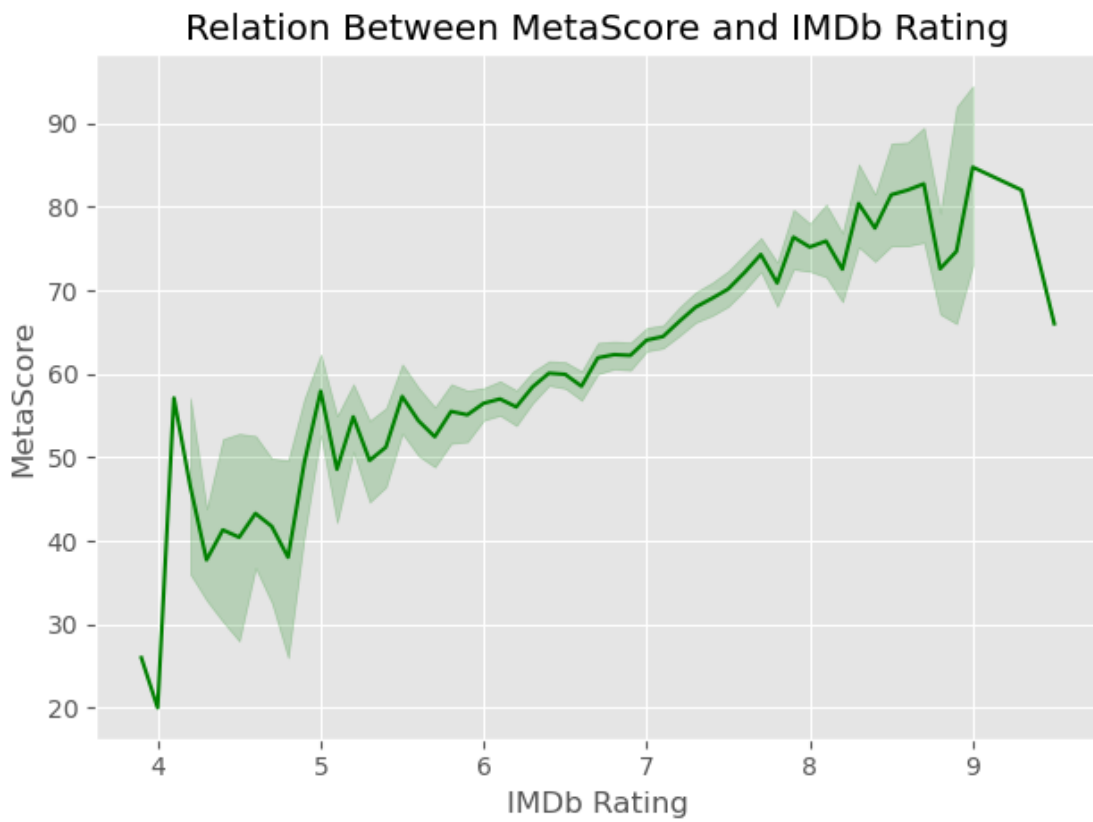
[28]:
```
plt.style.use('ggplot')
```

[29]:
```
RatingTime = df.sort_values(by="Duration (minutes)",ascending =False)
sns.lineplot(data=RatingTime,y="Duration (minutes)",x="IMDb Rating")
plt.title("Relation Between Duration and IMDb Rating")
plt.tight_layout()
plt.show()
```



When we observe the line plot between duration and rating we can observe that audience are more intrested in the movies with duration around 120 - 170 minutes and not at all intrested to watch the show if it is 80 - 100 minutes

[31]:
```
IMDbMeta = df.sort_values(by="MetaScore",ascending =False)
sns.lineplot(data=RatingTime,y="MetaScore",x="IMDb Rating",color="green")
plt.title("Relation Between MetaScore and IMDb Rating")
plt.tight_layout()
plt.show()
```

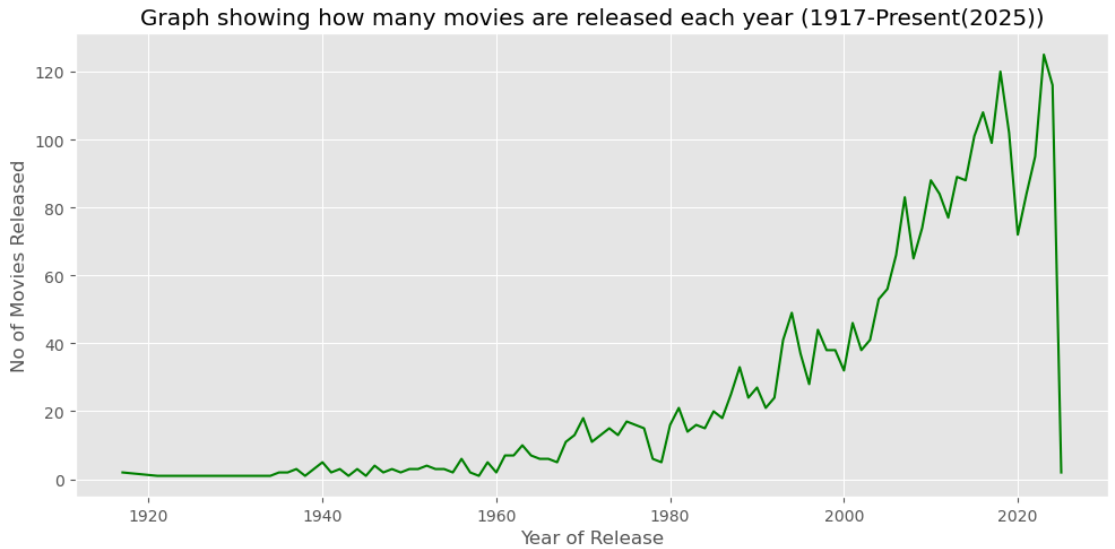


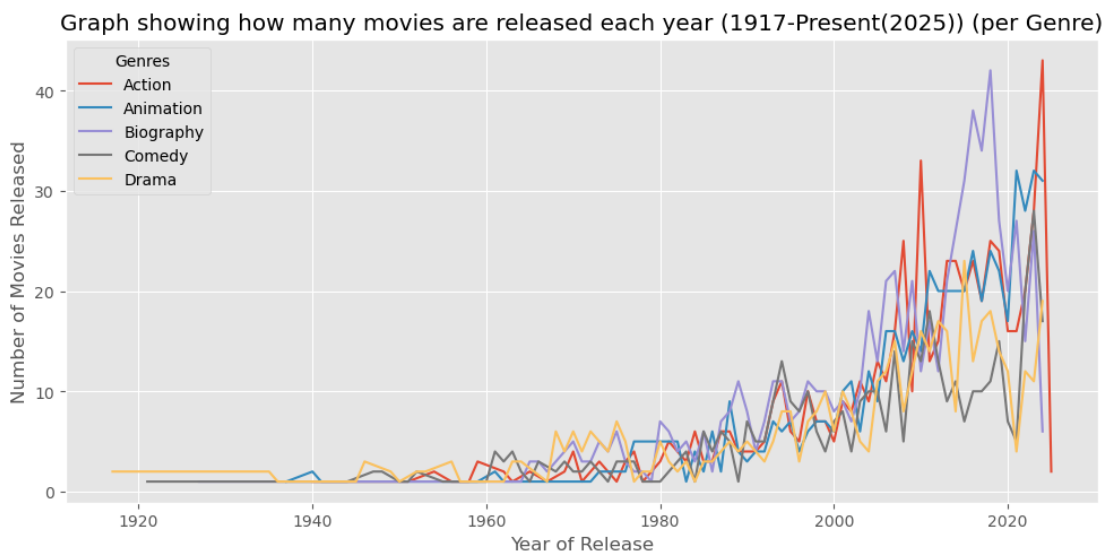
There is a clear case of Direct proportionality between IMDb and MetaScore

but after 9 IMDb rating people may be felt that ratings are overrated and the metascore fell

[33]:
```
fig, axes = plt.subplots(2,1 , figsize=(10, 10))
Year = df.groupby("Year")["Year"].count()
sns.lineplot(data=Year,ax=axes[0],color="green")
axes[0].set_xlabel("Year of Release")
axes[0].set_ylabel("No of Movies Released")
axes[0].set_title("Graph showing how many movies are released each year␣
↪(1917-Present(2025))")

genre_data = df.groupby(["Year", "Genre"])["Genre"].count().unstack()
for genre in ["Action", "Animation", "Biography", "Comedy", "Drama"]:
    sns.lineplot(data=genre_data[genre],label=genre)
axes[1].set_xlabel("Year of Release")
axes[1].set_ylabel("Number of Movies Released")
axes[1].set_title("Graph showing how many movies are released each year␣
↪(1917-Present(2025)) (per Genre)")
axes[1].legend(title="Genres")
```

```
plt.tight_layout()
plt.show()
```





The First Graph shows the significant increase of films from 1980 to present . And the peak of these genres are noted in the year 2023

If we observe clearly we can also observe the fall in the number of films released at the time of pandemic

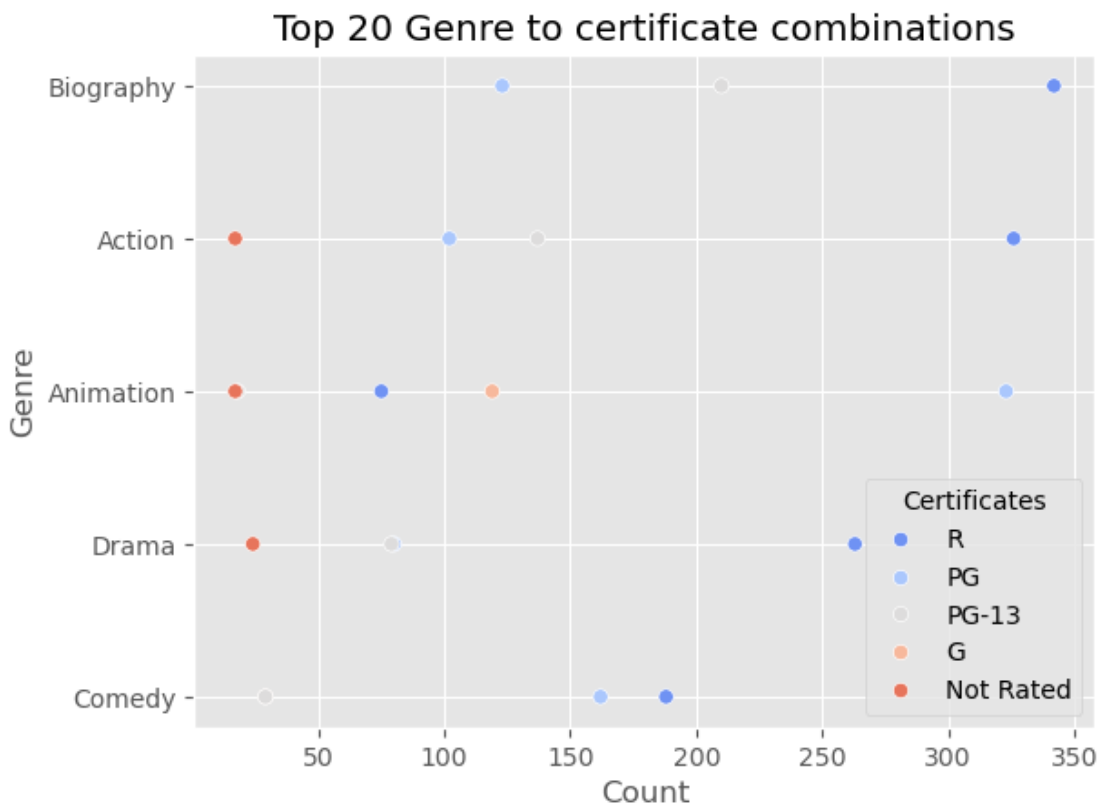
The Second Graph Shows that the increase in Biography was too high upto 2018-19 but after that people began to encourage Action and Animation. Comedy and Drame have been consistent since 1990s

[35]:
```
GenCert = df.groupby(["Genre"])["Certificates"].value_counts().
 ↪reset_index(name="Count").sort_values(by="Count",ascending = False).head(20)
GenCert
```

[35]:

	Genre	Certificates	Count
22	Biography	R	342
0	Action	R	326
11	Animation	PG	323
48	Drama	R	263
23	Biography	PG-13	210
34	Comedy	R	188
35	Comedy	PG	162
1	Action	PG-13	137
24	Biography	PG	123
12	Animation	G	119
2	Action	PG	102
49	Drama	PG	80
50	Drama	PG-13	79
13	Animation	R	75
36	Comedy	G	29
37	Comedy	PG-13	29
51	Drama	Not Rated	24
14	Animation	PG-13	18
3	Action	Not Rated	17
15	Animation	Not Rated	17

[36]:
```
sns.
 ↪scatterplot(data=GenCert,x="Count",y="Genre",hue="Certificates",palette="coolwarm",legend=T
plt.title("Top 20 Genre to certificate combinations")
plt.show()
```



We can observe the most crazy combinations are Biography-R rated , Action-R Rated, Animation-PG Rated

1.4 Stars who acted in the most number of movies

[39]:
```
df1 = pd.concat(map(pd.read_csv, ['IMDb_movies/IMDb_Dataset_1.csv',
                                  'IMDb_movies/IMDb_Dataset_2.csv',
                                  'IMDb_movies/IMDb_Dataset_3.csv']),␣
 ↪ignore_index=True)
```

[40]:
```
df1 = df1.drop_duplicates()
df = df1[(df1["Genre"] == "Action") |
         (df1["Genre"] == "Comedy") |
         (df1["Genre"] == "Animation") |
         (df1["Genre"] == "Biography") |
         (df1["Genre"] == "Drama")]
```

[41]:
```
import re
df.loc[:, "Star Cast"] = df["Star Cast"].str.replace(r'(?<=[a-z])(?=[A-Z])', '␣
 ↪, ', regex=True)
```

[42]:
```
df.loc[:, 'Star Cast'] = df['Star Cast'].astype('string').str.split(' , ')
StarCast = df.explode('Star Cast').drop_duplicates()
StarCast
```

[42]:
```
                                          Title  IMDb Rating  Year  \
0                                     Gladiator          8.5  2000
0                                     Gladiator          8.5  2000
0                                     Gladiator          8.5  2000
1     Mission: Impossible - Dead Reckoning Part One          7.7  2023
1     Mission: Impossible - Dead Reckoning Part One          7.7  2023
…                                             …          …   …
4568                         The Boy and the Heron          7.5  2023
4569                               Despicable Me 4          6.3  2024
4570                                    Inside Out          8.1  2015
4571                                            IF          6.5  2024
4572                                  Inside Out 2          7.9  2024

     Certificates      Genre               Director          Star Cast  \
0               R     Action           Ridley Scott     David Franzoni
0               R     Action           Ridley Scott         John Logan
0               R     Action           Ridley Scott  William Nicholson
1           PG-13     Action  Christopher McQuarrie         Tom Cruise
1           PG-13     Action  Christopher McQuarrie      Hayley Atwell
…           …        …                     …                 …
4568        PG-13  Animation         Hayao Miyazaki       Soma Santoki
4569           PG  Animation           Chris Renaud       Steve Carell
4570           PG  Animation            Pete Docter        Amy Poehler
4571           PG  Animation         John Krasinski     Cailey Fleming
4572           PG  Animation            Kelsey Mann        Amy Poehler

      MetaScore  Duration (minutes)
0          67.0               155.0
0          67.0               155.0
0          67.0               155.0
1          81.0               163.0
1          81.0               163.0
…         …                  …
4568       91.0               124.0
4569       50.0                95.0
4570       94.0                95.0
4571       46.0               104.0
4572       73.0                96.0

[6782 rows x 9 columns]
```

Note : The above cell is to be run only once if it is run more than once it will remove all star cast and we get no output

In case the cell is already run twice and showing NaN in Star Cast please run the df declaration cell again

[44]:
```
CastDir = StarCast.groupby(["Director","Star Cast"])["Star Cast"].count().
 ↪reset_index(name="Count")
CastDir = CastDir[CastDir["Director"] != CastDir["Star Cast"]]
CastDir.sort_values(by="Count",ascending=False).head(10)
```

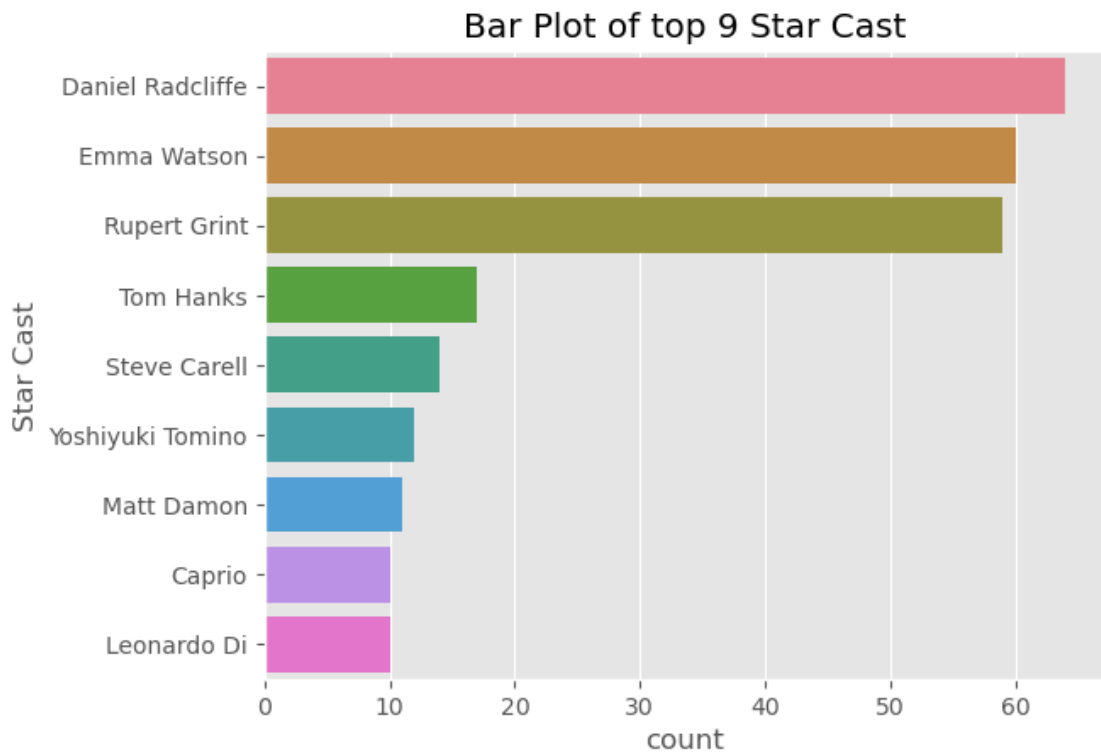
[44]:

	Director	Star Cast	Count
5851	Steven Spielberg	Tom Hanks	4
1973	Genndy Tartakovsky	Adam Sandler	3
4690	Peter Jackson	J.R.R. Tolkien	3
635	Bill Corbett	Rupert Grint	3
634	Bill Corbett	Emma Watson	3
633	Bill Corbett	Daniel Radcliffe	3
2642	Jeff Fowler	Ben Schwartz	3
282	Andrew Erwin	Jon Erwin	3
1461	Dean DeBlois	Jay Baruchel	3
6479	Zack Snyder	Shay Hatten	3

These are the top 10 times a Director made repeated film with star cast

The Data above does not include the count where director is same in star cast

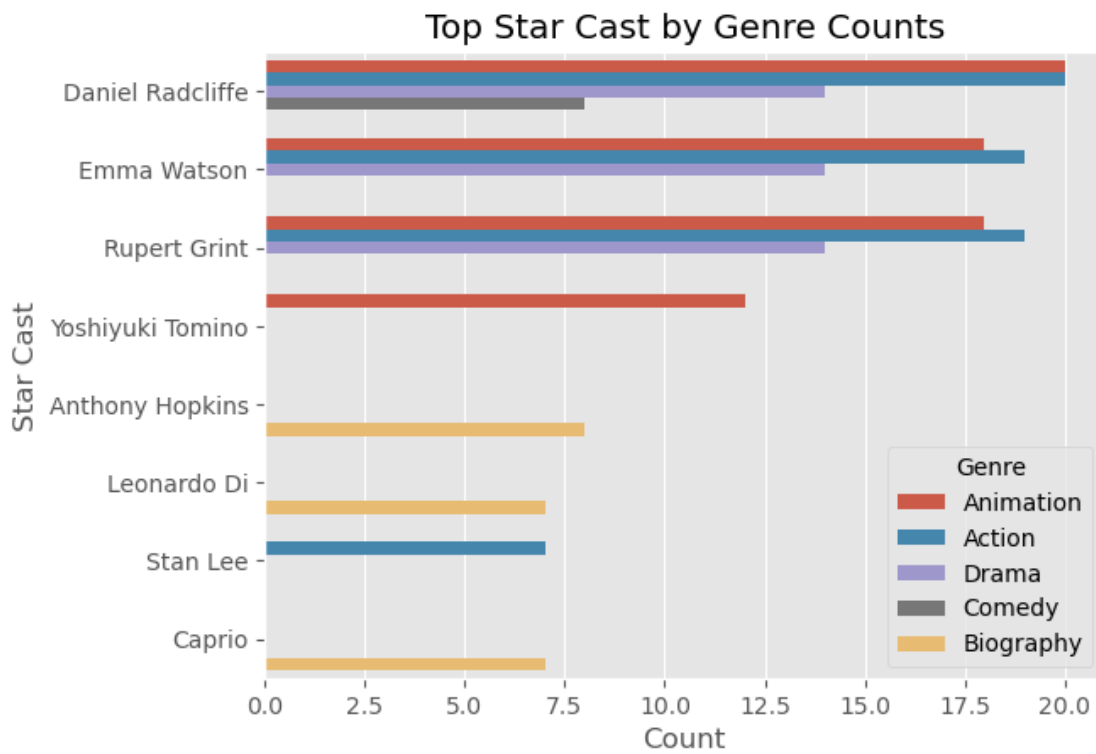
[113]:
```
StarMov = StarCast["Star Cast"].value_counts().head(9).reset_index(name="count")
sns.barplot(data=StarMov,x="count",y="Star Cast",hue="Star Cast")
plt.title("Bar Plot of top 9 Star Cast")
plt.show()
```



From my selected genres : Harry Potter Movie Series cast(especially Daniel Radcliffe ,Emma Watson ,Rupert Grint) has got the most number of movies in the data frame

[48]:
```
StarGen = StarCast.groupby("Star Cast")["Genre"].value_counts()
StarGen = StarGen.sort_values(ascending=False).head(15).
 ↪reset_index(name="Count")
```

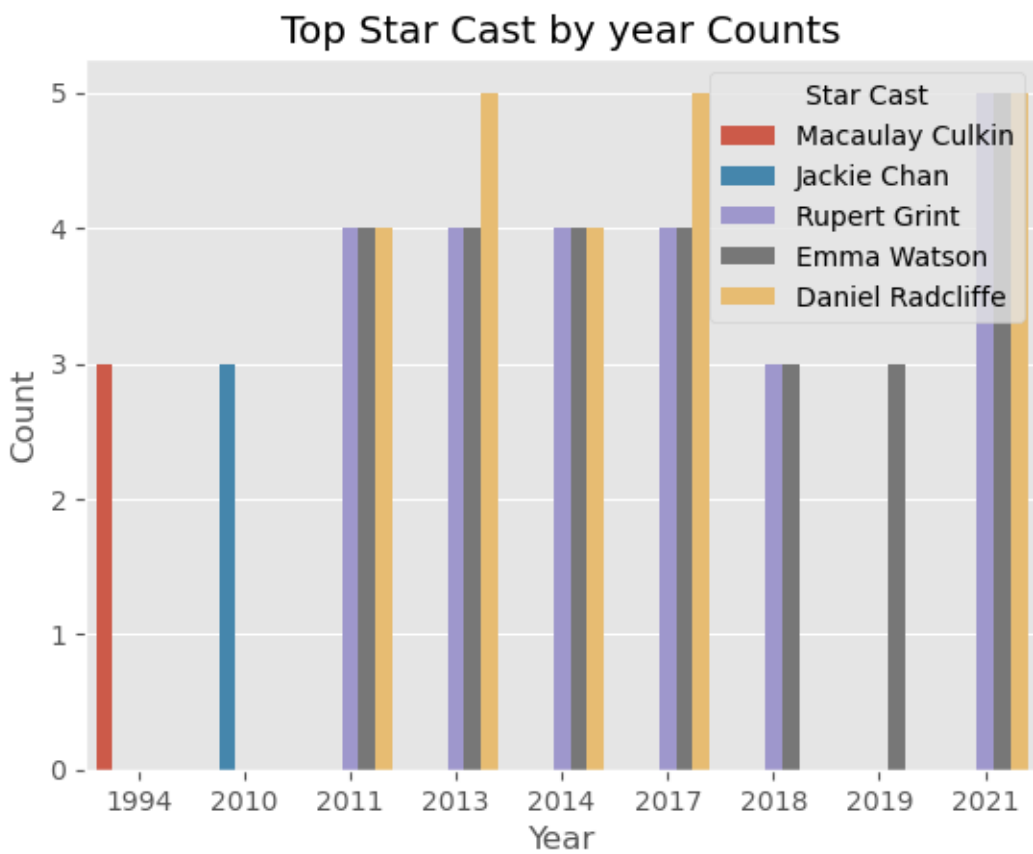
[49]:
```
sns.barplot(data=StarGen, y="Star Cast", x="Count", hue="Genre", orient="h")
plt.title("Top Star Cast by Genre Counts")
plt.show()
```



This Bar graph between Cast and movie count with respect to Genre clarifies that harry potter cast due to Action , Animation and Drama in their genre they got many movie names count

[51]:
```
StarYr = StarCast.groupby("Year")["Star Cast"].value_counts()
StarYr = StarYr.reset_index(name="Count").
 ↪sort_values(by="Count",ascending=False).head(20)
```

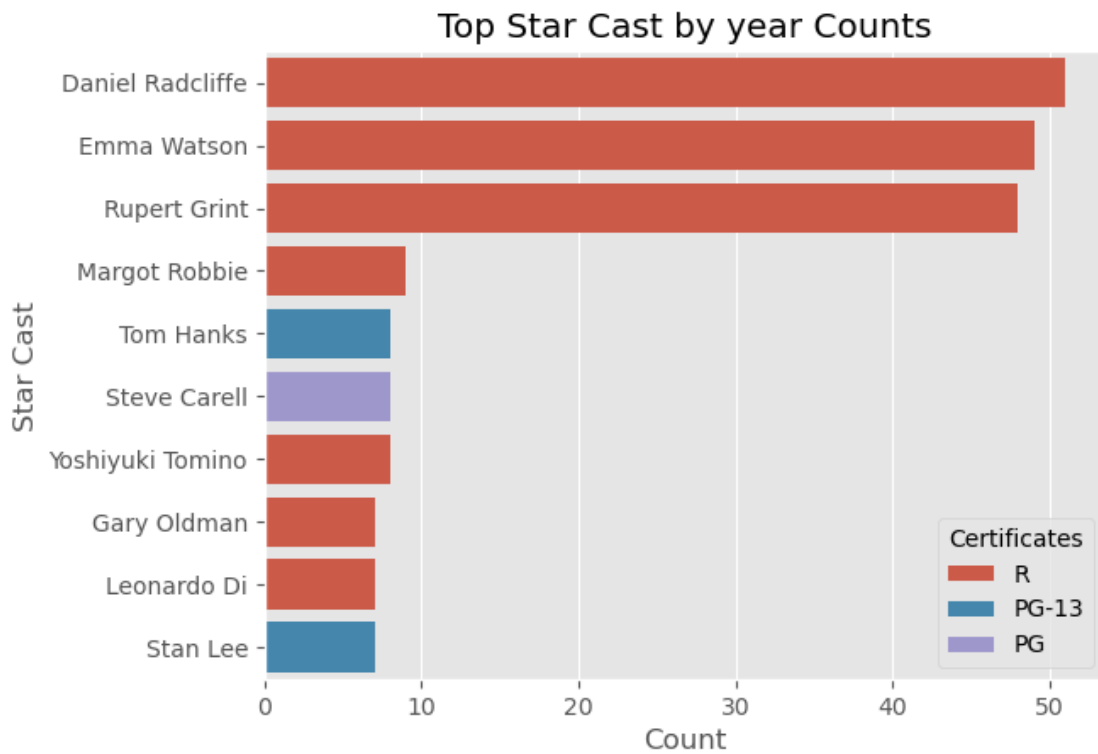
[52]:
```
sns.barplot(data=StarYr, y="Count", x="Year", hue="Star Cast",legend=True)
plt.title("Top Star Cast by year Counts")
plt.show()
```



Even from this graph we can interpret that Harry potter cast (Daniel Radcliffe ,Emma Watson ,Rupert Grint) has the most number of movies track record followed by Jackie Chan and Macaulay Culkin

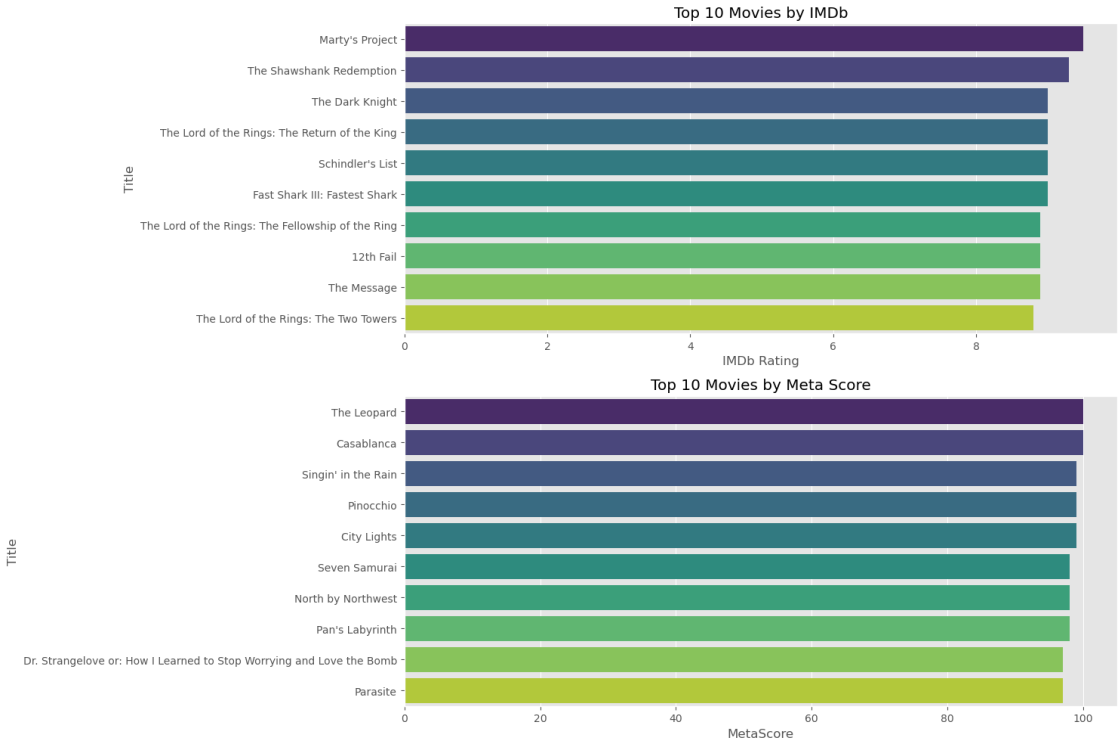
[54]:
```
StarCert = StarCast.groupby("Certificates")["Star Cast"].value_counts()
StarCert = StarCert.reset_index(name="Count").
 ↪sort_values(by="Count",ascending=False).head(10)
```

[55]:
```
sns.barplot(data=StarCert, y="Star Cast", x="Count", hue="Certificates")
plt.title("Top Star Cast by year Counts")
plt.show()
```



The Genres I have chosen (Action, Comedy, Animation, Biography, Drama) seems like mostly Dominated by this trio (Daniel Radcliffe ,Emma Watson ,Rupert Grint) (Indicating large R rated certifications)

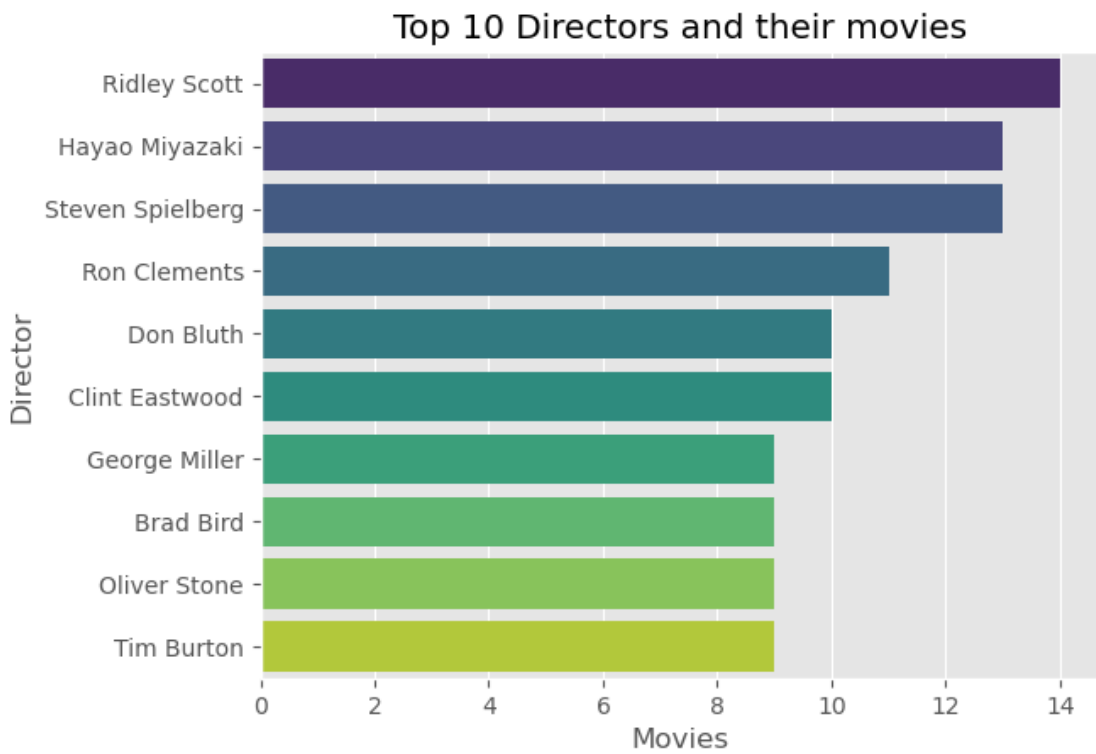
[57]:
```
fig, axes = plt.subplots(2 ,1 , figsize=(15, 10))
DFIMDb = df.sort_values(by="IMDb Rating",ascending = False).head(10)
sns.barplot(data =DFIMDb , y= "Title",x="IMDb Rating", ax =␣
 ↪axes[0],hue="Title",palette ="viridis")
axes[0].set_title("Top 10 Movies by IMDb")
MetaDF = df.sort_values(by="MetaScore",ascending = False).head(10)
sns.barplot(data =MetaDF , y= "Title",x="MetaScore", ax =␣
 ↪axes[1],hue="Title",palette="viridis")
axes[1].set_title("Top 10 Movies by Meta Score")
plt.tight_layout()
plt.show()
```



The first bar graph shows the top 10 movies by IMDb and second graph shows the audience intrest and are suggested to watch

1.4.1 analysing Top Directors performance

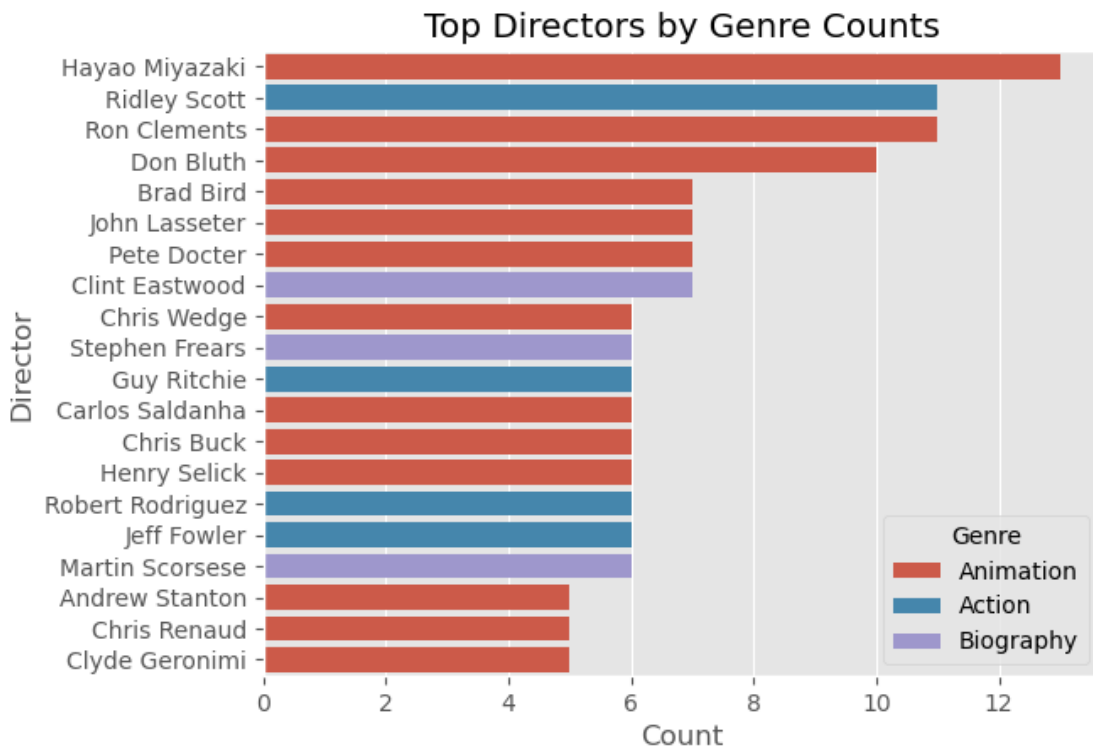
[60]:
```
DirMov = df.groupby(["Director"])["Title"].count().reset_index(name="Movies").
 ↪sort_values(by="Movies",ascending = False).head(10)
sns.
 ↪barplot(data=DirMov,x="Movies",y="Director",hue="Director",palette="viridis",legend=False)
plt.title("Top 10 Directors and their movies")
plt.show()
```



This Plot names the top 10 directors(According to Number of movies) and we observe That Ridley Scott (Director of Gladiator) has the most number of movies(14)

[62]:
```
DirGenre = df.groupby(["Genre"])["Director"].value_counts().reset_index(name␣
↪="Count").sort_values(by="Count",ascending =False).head(20)
```

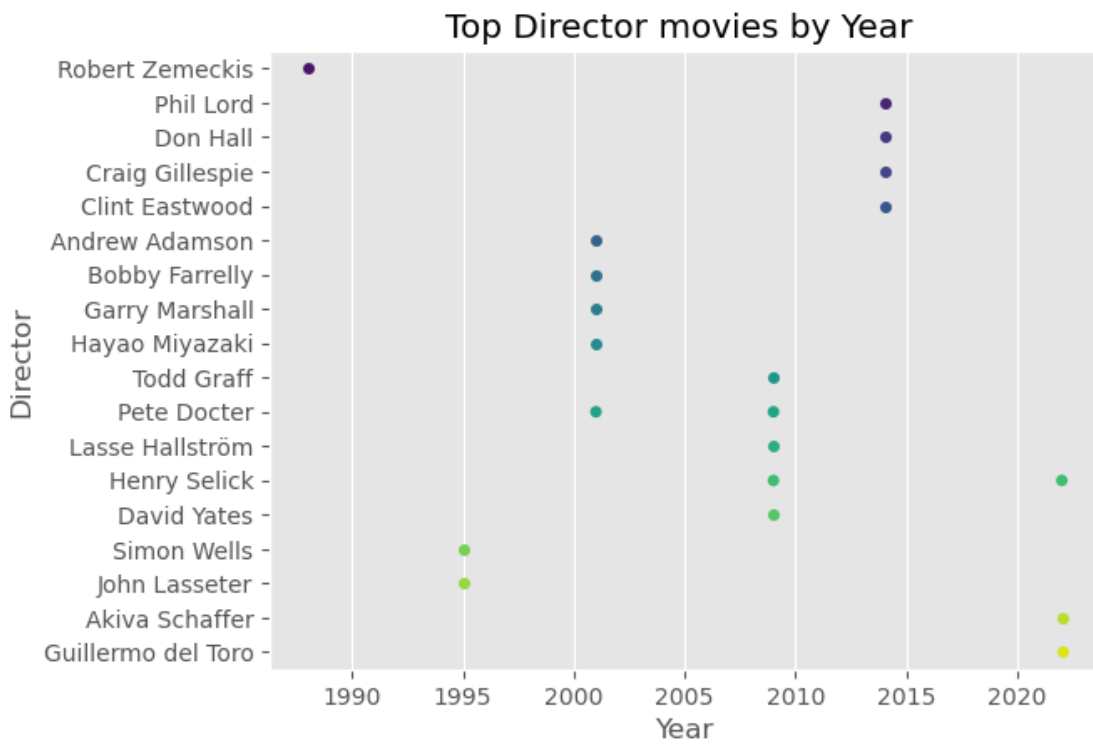
[63]:
```
sns.barplot(data=DirGenre, y="Director", x="Count", hue="Genre", orient="h")
plt.title("Top Directors by Genre Counts")
plt.show()
```



From this graph we can interpret that many directors have numerous movies in single genre and visually I can say that most of them are fond of animation genre

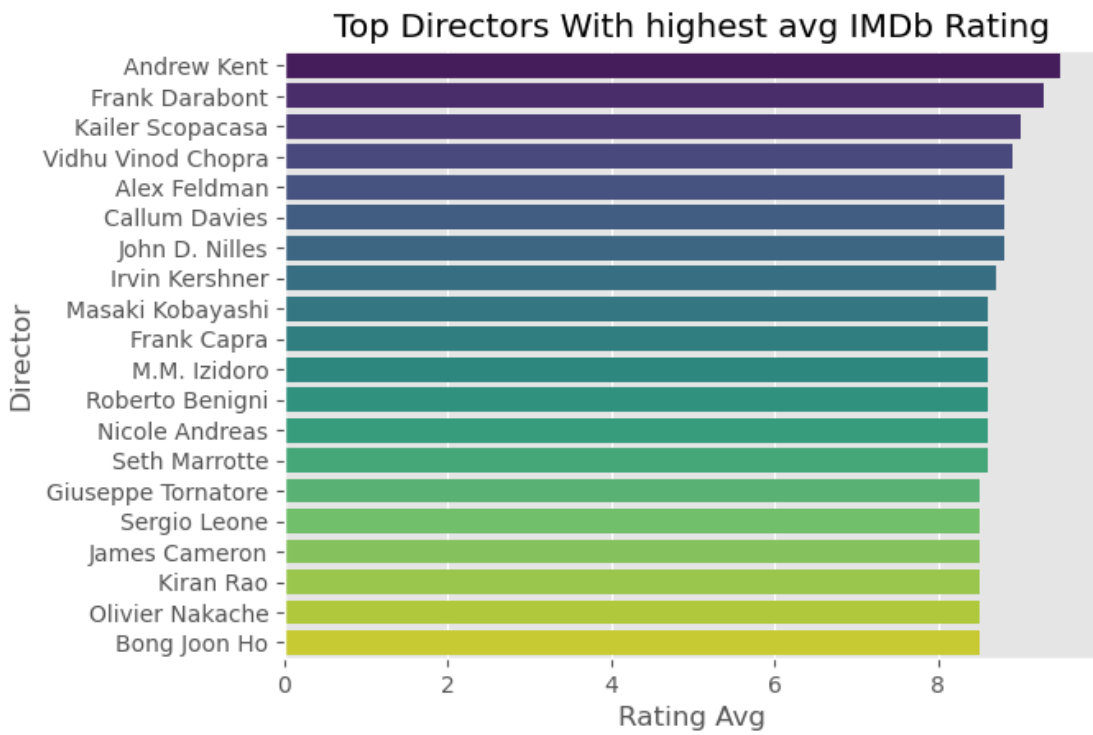
[65]:
```
DirYr = df.groupby(["Year"])["Director"].value_counts().reset_index(name␣
↪="Count").sort_values(by="Count",ascending =False).head(20)
```

[66]:
```
sns.swarmplot(data=DirYr, y="Director", x="Year",␣
↪hue="Director",palette="viridis", orient="h")
plt.title("Top Director movies by Year ")
plt.show()
```



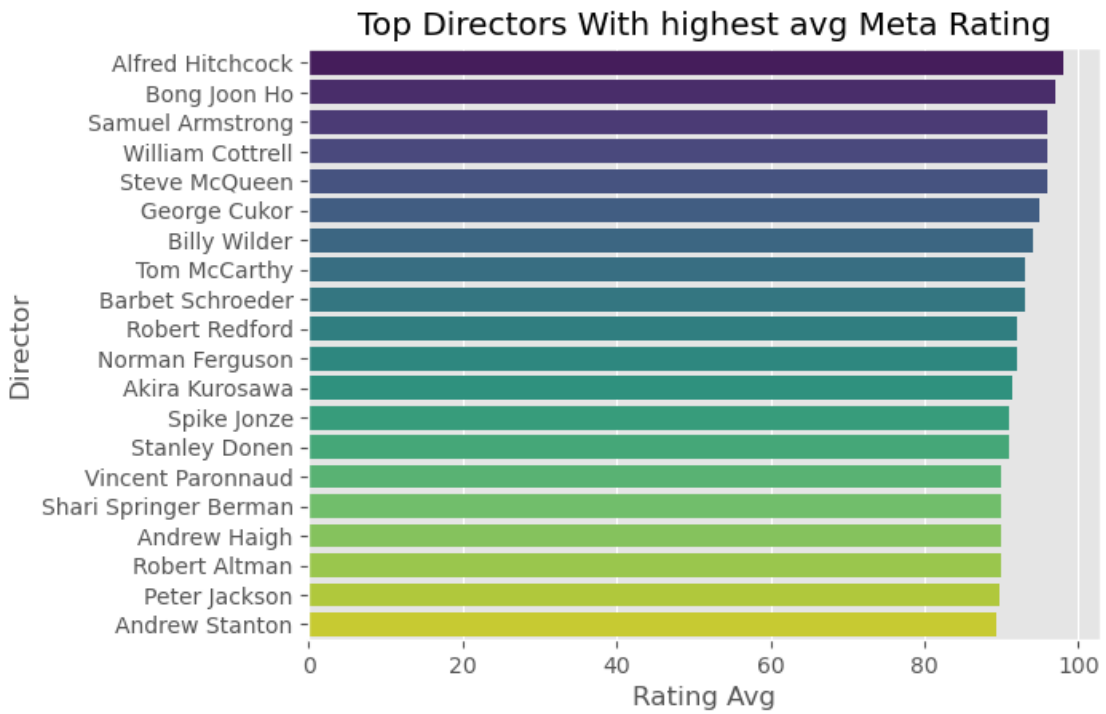
The Above scatter plot shows that there are many directors who have released 2 films a year but no one has released 3 films in a year

[68]:
```
DirAvg = df.groupby("Director")["IMDb Rating"].mean().reset_index(name ="Rating␣
↪Avg").sort_values(by="Rating Avg",ascending =False).head(20)
sns.barplot(data=DirAvg , x="Rating Avg",␣
↪y="Director",hue="Director",legend=False,palette="viridis")
plt.title("Top Directors With highest avg IMDb Rating")
plt.show()
```



These are the Top 20 directors suggested by IMBd rating

[70]:
```
DirFav = df.groupby("Director")["MetaScore"].mean().reset_index(name ="Rating␣
↪Avg").sort_values(by="Rating Avg",ascending =False).head(20)
sns.barplot(data=DirFav , x="Rating Avg",␣
↪y="Director",hue="Director",legend=False,palette="viridis")
plt.title("Top Directors With highest avg Meta Rating")
plt.show()
```



These are top 20 directors suggested by Meta Rating

1.4.2 Conclusion

Genre and Certification Trends: Biography and Action dominate genres, while PG-13 certification is most common, reflecting a focus on family-friendly content.

Star Cast and Directors: Franchise actors like Daniel Radcliffe and directors such as Alfred Hitchcock consistently perform well in terms of both quantity and quality.

Ratings and Scores: Longer movies tend to have higher IMDb ratings, while critically acclaimed classics lead MetaScores.

Yearly Trends: Movie releases surged post-2000 due to technological advancements and streaming platforms but declined recently, likely due to external factors or incomplete data.